

# ExonViz: Transcript visualization made easy

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## Software

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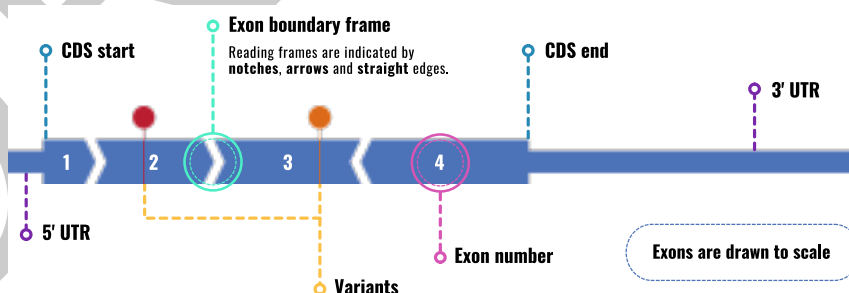
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## 9 Summary

Transcripts of a gene contain one or more **exons**, which encode the functional parts of the transcript, and **introns**, which are removed in a process called **splicing**. A single gene typically encodes multiple **transcripts** by including different exons. Protein coding genes include one or more coding exons, which encode a protein using three-letter sequences called **codons**. These codons do not necessarily coincide with exon boundaries, because a single codon can span two exons. If the codon boundaries of adjacent exons are not aligned, an often detrimental **frame shift** is introduced. Taking exon boundary reading frames, which we will call **exon boundary frames**, into account is crucial when considering the effect of mutations and when designing genetic therapies such as exon skipping.

19 ExonViz is a Python package and web application that creates biologically accurate RNA  
20 transcript figures, including features such as coding regions, genetic variants and exon boundary  
21 frames. Any transcript defined by Ensembl or RefSeq (which contains over 150.000 species as  
22 of 2026) can be retrieved and visualized automatically.



**Figure 1:** Example transcript highlighting ExonViz features. **5' UTR:** Non coding region at the start of the transcript. **CDS start:** Start of the coding region. **CDS end:** End of the coding region. **3' UTR:** Non coding region at the end of the transcript.

## 23 Statement of need

Visualization of transcripts, including features like exon boundary frames, coding and non coding regions is important within the field of clinical and human genetics (Walker et al., 2023). Illustrating the exon structure and the location of variants is common practice, especially when new genes, variants or transcripts have been discovered. These illustrations are also used to assess potential genetic treatment options (e.g., canonical exon skipping), in teaching

29 settings, in diagnostics, to identify mutational hotspots and for genetic counseling. In particular,  
30 knowledge about the exon boundary frames aids in the assessment of the pathogenicity of  
31 genetic variants using the ACMG-AMP guidelines (Richards et al., 2015), when evaluating exon  
32 spanning deletions (Cheerie et al., 2025) and when interpreting the effects of splice altering  
33 variants (Walker et al., 2023).

34 To date, most people have to resort to manually drawing transcripts with tools like Illustrator,  
35 Photoshop or BioRender, or forgo illustrations altogether. Creating transcript visualizations  
36 should be quick and easy to be utilized in clinical and day to day settings, rather than to create  
37 a bespoke figure for a manuscript or presentation.

## 38 State of the field

39 Several tools have been made available to visualize various aspects of genes and transcripts.  
40 ggtranscript (Gustavsson et al., 2022) and wiggleplotr (Alasoo, 2017) can visualize transcripts  
41 and exons, while tools like genepainter (Mühlhausen et al., 2015) or Swan (Reese & Mortazavi,  
42 2021) can be used to visualize different transcript isoforms. Variants can be shown on the  
43 transcript with Variant View (Ferstay et al., 2013). However, all of these tools require  
44 substantial expertise to setup and retrieve the required transcript models, which make them  
45 hard to use for users with minimal technical expertise. Furthermore, none of them have the  
46 option to indicate exon boundary frames.

47 To our knowledge there are currently no easily usable tools available which allow a non-technical  
48 user to quickly draw all features required for a comprehensive overview of a transcript's structure  
49 and the location of variants of interest.

## 50 Software design

51 Biological transcripts are deceptively complex and can consist of introns, exons and a coding  
52 region defined on either the forward or reverse strand of a reference sequence. Introns are  
53 usually the largest structures in a transcript, as a single intron can be larger than all exons of  
54 a transcript combined.

55 For ExonViz, we opted to hide this complexity and focus on the exon as the central unit,  
56 with a transcript being defined simply as a sequence of Exons. We also ignore the strand  
57 of the transcript on the reference sequence, effectively rewriting all transcripts to be defined  
58 on the forward orientation. The information of each exon is stored in the Exon class, which  
59 consist only of a size and optional coding region, without a specified location on the reference  
60 sequence. Exons can also have a name, color and a list of (named) variants which will be  
61 included in the visualization.

62 The Exon class also hold the required functionality for drawing an exon, such as the rendered  
63 size of the exon, the minimum scale at which the exon can be drawn, as well as methods to  
64 split an Exon (analogous to how long words can be split over multiple lines).

65 The Exon class can be imported and used directly and users can also specify exons in a simple  
66 TSV format as described in the documentation of ExonViz. To give users access to existing  
67 transcript definitions, ExonViz supports automatic retrieval of transcript definitions from the  
68 Mutalyzer API (Lefter et al., 2021), which helps avoid the complexities of retrieving and  
69 processing various different transcript definition formats. This gives ExonViz access to all  
70 transcripts defined in the RefSeq (O'Leary et al., 2016) and Ensembl (Harrison et al., 2023)  
71 databases across many species, ranging from human and mouse to fruit fly and coelacanth.  
72 This wide range of supported transcripts is also why we decided not to bundle any transcript  
73 definitions with ExonViz itself.

74 Mutalyzer is actively being developed in our group and will be maintained for the foreseeable

75 future, which made it an obvious choice to use. Should the mutalyzer website become  
76 unavailable, it is also possible to run a local installation of mutalyzer with caching for offline  
77 usage.

78 Due to the simple structure of the Exon class, adding novel sources of transcript information  
79 in the future, such as gtf or gff files, should not be too difficult.

## 80 Method

81 ExonViz visualizes the exon boundary frames by using different shapes for the boundary of  
82 exons. Figure 2 shows all possible combinations of exon and codon boundaries, and the  
83 corresponding exon boundary shapes. When the exon and codon boundaries coincide (frame  
84 0) the exons are drawn with a straight edge, as is the case of exon 1 and 2. Exon 2 ends one  
85 base into the codon (in frame 1), which is drawn using an arrow on the end of the exon. Exon  
86 3 starts in frame 1, and is drawn with a notch at the start of the exon. This is reversed for the  
87 boundary between exons 3 and 4, which is in frame 2. Since the exons of a transcript should  
88 fit together, exons in conflicting frames (e.g. because of a frame shift inducing variant) are  
89 easily spotted due to the fact that the exon boundaries do not fit together.



**Figure 2:** Visualization of the relation between codons and exon boundary frames. The shapes of the exons illustrate the relation between the exon boundaries and the codon boundaries.

90 The output of ExonViz is an SVG figure generated using the svg-py library, which can be used  
91 directly or modified using modern graphical editing programs. It is also possible to output the  
92 transcript and variants in TSV format, edit the transcript using any text editor or spreadsheet  
93 program, and draw the modified transcript using ExonViz. The [online documentation](#) has a  
94 number of examples of custom transcripts that can be visualized this way.

## 95 Research impact statement

96 ExonViz has proven to be a useful resource to quickly visualize exon reading frames and  
97 check the location of variants in a transcript. ExonViz is actively being used in the field  
98 of personalized medicine and is one of the recommended resources in the latest consensus  
99 guidelines for assessing pathogenic variants for RNA therapies (Cheerie et al., 2025). In  
100 addition, the [ExonViz website](#) has been used to generate over 14000 transcript figures between  
101 September 2023 and August 2026.

## 102 Conclusion

103 To our knowledge, ExonViz is the first publicly available application that allows for automatic  
104 visualization of transcripts with additional features such as exon boundary frames and variants  
105 along the transcript. ExonViz can be used for illustrations within publications, assessment of  
106 treatment options, teaching purposes and genetic counseling. Figures generated by ExonViz  
107 are free to use under the Creative Commons BY license. Furthermore, we allow the user to  
108 construct their own transcripts, for example to visualize non-standard exons or alternative  
109 isoforms. ExonViz can be accessed as a web application via [exonviz.rnatherapy.nl](https://exonviz.rnatherapy.nl) or installed  
110 via [PyPI](#). The source code is available on [Github](#).

## 111 AI usage disclosure

112 No generative AI tools have been used for ExonViz, the documentation or this manuscript.

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